QR-011 Fusion add-in: planning stock

Doc ID	QR-011
Title	Fusion add-in: planning stock
Parent	CS-003 §5; CS-003 §7.1; CS-003 §7.2; CS-003 §7.3
Checked against	CS-003 Rev E; CS-003 Rev E; CS-003 Rev E; CS-003 Rev E

Derived from CS-003. If this and the parent disagree, the parent is right. Install and troubleshooting steps are copied from `10 Fusion Add-in/FEBPlanStock/README.md`, `10 Fusion Add-in/installer/INSTALL.txt` and `06 Composites App/app/fusion.js`.

1. How do I install the Fusion add-in?

On the app's Molds tab, press Plan from Fusion. It opens the latest release page. Download `FEBPlanStock-<version>.zip`, unzip it, double-click the installer for your computer, then quit Fusion and open it again (CS-003 §5; add-in README).

Also called: Fusion, Fusion 360, add-in, addin, plugin, plug-in, FEBPlanStock, Plan stock, install, installer.

1. In the app, open the Molds tab and press Plan from Fusion.
2. Download `FEBPlanStock-<version>.zip` from the release page.
3. Unzip the whole download. The installer has to see the FEBPlanStock folder next to it.
4. On a Mac, double-click `Install on Mac.command`. On Windows, double-click `Install on Windows.bat`.
5. Quit Fusion and open it again.
6. Check that the FEB panel is on the Utilities tab of the Design workspace, with a Plan stock button on it.

To load it without restarting: Utilities, Add-Ins, the Add-Ins tab, select FEBPlanStock, press Run (add-in README).

On first use the panel shows the app's sign-in card. Sign in with your own composites app account. It remembers you on that machine, and your name is stamped on the molds you plan (INSTALL.txt).

The Windows installer follows Autodesk's documented path but had not been run on a Windows machine when it shipped. If you install on Windows, tell the composites lead how it goes (INSTALL.txt).

To update, download the newer zip and run the installer again. It replaces the add-in and keeps your settings (INSTALL.txt).

2. macOS says "Apple could not verify it is free of malware"

That is expected the first time you run the installer from a download, and it clears in three steps: System Settings → Privacy & Security → Open Anyway, then double-click the installer again and press Open (INSTALL.txt).

Also called: Mac security warning, Gatekeeper, unidentified developer, cannot be opened, malware warning, Open Anyway, quarantine.

1. Press Done on the dialog.
2. Open System Settings, then Privacy & Security.
3. Scroll down to Security. Find the line about "Install on Mac.command" and press Open Anyway.
4. Double-click the installer again.
5. Press Open.

It happens once per download and does not come back (add-in README). The team does not pay Apple to sign the installer, which is why the dialog appears (INSTALL.txt).

The installer also clears the quarantine flag a browser puts on every file in a downloaded zip, so the add-in itself loads without a second warning (add-in README).

Do not copy the add-in folder in by hand to get round the dialog. Copying by hand overwrites `credentials.json`, which the installer keeps (add-in README).

3. Planning a mold's stock from Fusion, step by step

Open the mold design, press Utilities tab → FEB panel → Plan stock, select the Mold body (one body at a time), set density in the app's mold dialog that opens in the palette, and press Plan (CS-003 §7.1 step 9).

Also called: plan stock, stock plan, stack plan, board plan, plan from Fusion, mold body, bottom face, CAM stock.

1. Open the mold design in Fusion.
2. Press Utilities tab → FEB panel → Plan stock.
3. Select the Mold body. Plan one body at a time (CS-003 §7.1 step 8).
4. If the mold is not modelled bottom-down (a split mold on its side), also select the Bottom face: the flat face the stack is glued up from (CS-003 §7.1 step 9).
5. Press OK. The FEB Composites palette opens on the right with the mesh loaded.
6. Sign in if the palette asks. The mold opens by itself once you have.
7. In the mold dialog, set the board density. For a range, use Board density, min (lb/ft³) and Board density, max (lb/ft³) (CS-003 §5).
8. Press Plan.
9. Save the Fusion document yourself. The add-in saves nothing to it (CS-003 §7.1 step 9).

Grades shall not be swapped silently: plan at one grade, or give the range in the dialog (CS-003 §5).

With Boards set to choose them for me, from stock (the default), the planner picks thicknesses only from boards recorded in Inventory → Boards. I'll pick the thicknesses is not checked against the rack: use it only for boards you have in hand. Either way it splits any stack over 6 in into sections (CS-003 §7.1 step 8). If no board of your density is recorded, add boards, widen the range, or plan at a density that is on the rack (add-in README).

Plan the stack before the Mold design review (CS-003 §7.1 step 8).

4. Where does the plan go after I press Plan?

Into the app. The planner creates the mold record at Designed with the plan (`STK-SN6-###`) as its Mold file, and Fusion draws the blanks as see-through bodies in a component named after the plan. Use them as CAM stock (CS-003 §7.1 steps 8 and 9).

Also called: STK, stack plan, Mold file, where did my plan go, blanks, blocks, stock bodies, component.

Where	What you find
Fusion	A component named after the plan (`STK-SN6-###`), one see-through body per blank. Fusion confirms "[plan] is saved in the app … Nothing is saved to this document yet."
App, Molds tab	The new mold record (`MOLD-SN6-###`) at stage Designed
App, the mold's page	The plan under Mold file, and the Stack drawing (under the heading Stack, below Mold in stock)

Where	What you find
App, Molds → Cut list	The blanks to cut. The cut list does not check the review: cut only for a mold whose Mold design review is signed (CS-003 §7.2; §7.3 step 1)

Running Plan stock again on the same mold replaces that component in Fusion (add-in README).

STL planning is marked beta. Check the Stack drawing on the mold page (under the heading Stack) before anyone cuts, and plan from dimensions if it looks wrong (CS-003 §7.1 step 8).

Re-plans are a lead's. On a mold that already has a plan, a lead presses Re-plan; the old plan stays under Earlier, marked Superseded. Cut and machine only from the current plan (CS-003 §7.1 step 8).

A split mold's pieces are separate mold records. Link every piece on the part page with + Link a mold…, then + Link another mold…. Each piece gets its own run and its own design review. + New run fills in the part's first linked mold: on the second piece's run, set Mold ID in the run's Details to that piece's `MOLD-SN6-###` (CS-003 §7.1 step 10).

5. The app says my Fusion add-in is out of date

The app found an add-in older than it expects. Reinstall it: press Plan from Fusion on the Molds tab, download the newest zip, run the installer again, and restart Fusion (`app/fusion.js`; CS-003 §5).

Also called: out of date, old add-in, outdated, update the add-in, version mismatch, add-in version, reinstall.

The message reads: "This machine's Fusion add-in is [its version] and the app expects [the minimum]. Reinstall it from Plan from Fusion on the Molds tab." It shows once per page and never stops you: an old add-in usually still plans a mold, it just misses what the newer one learned to send (`app/fusion.js`).

1. On the Molds tab, press Plan from Fusion.
2. Download the newest `FEBPlanStock-<version>.zip`.
3. Unzip it and run the installer over the old install. That is how you update.
4. Quit Fusion and open it again.

The add-in carries the app's own version number, not one of its own. The number on the zip is the app version it was built against, so an add-in and an app showing the same number are a matched pair (add-in README).

An add-in old enough to send no version at all also counts as out of date (`app/fusion.js`).

6. What does the split waiver measure?

Nothing. Ticking "Machined depth stays under 6in" is a claim by the person ticking it. The app records who ticked it and when and shows the mold as split waived, but it does not measure the cavity (CS-003 §7.1 step 6; `app/molds.js`).

Also called: split waiver, split waived, no split, do not section, max-Z, max Z, cut depth, tall mold, 6 inch, sections.

The planner splits any stack over 6 in into sections automatically (CS-003 §7.1 step 6, `app/slicer.js`, 152.4 mm). The checkbox in the mold dialog stops that.

Tick it only when the design keeps the cutter within 6 in of a face it can reach (CS-003 §7.1 step 6):

- a shallow cavity in a tall blank;
- dowelled inserts;
- a face machined from both sides.

Leave it clear and a tall stack is split into sections, with the split's dowel and datum features designed in CAD, not improvised at the machine (CS-003 §7.1 step 6).

The design reviewer checks the claim. Mold design review check 6 asks for height within machine max-Z, or splits with alignment features designed, or a split waiver with the reviewer having checked the deepest machined point and the tool's reach past the collet (CS-003 §7.2). The machine's max-Z value is in CS-005 §5.

7. The add-in does nothing, or the mold never opens

Read `febplanstock.log` beside the installed add-in first: one line per step. If the palette says a mold from Fusion is waiting on something, do that thing (sign in, wait for the rack to load) and the mold opens by itself (add-in README; `app/fusion.js`).

Also called: not working, nothing happens, stuck, add-in broken, palette blank, log file, error, troubleshooting.

What you see	What it means and what to do
"A mold from Fusion is waiting on a sign-in."	Sign in in the palette. The mold opens once you have
"… waiting on a real sign-in (guests cannot create molds)"	Press Sign in and sign in with your own account
"… waiting on the rack to load"	Wait a few seconds; the board list is still loading
No board of your density on the rack	Add boards in Inventory → Boards, widen the range, or plan at a density that is recorded
A sign-in failure in a Fusion message box	Check `credentials.json`, or sign in in the palette yourself
Nothing after "sent mesh" in the log	The add-in reloads the page once. If that fails, close the palette, stop and run FEBPlanStock under Utilities, Add-Ins, and try again

If the palette is closed, Plan stock reopens it (add-in README).

If none of this helps, send `febplanstock.log` to the composites lead with what you were doing (INSTALL.txt).

To remove the add-in, delete the FEBPlanStock folder from Fusion's AddIns directory (INSTALL.txt).